Learning Human-Grounded Interaction Strategies through Action-Mediated Language Policies

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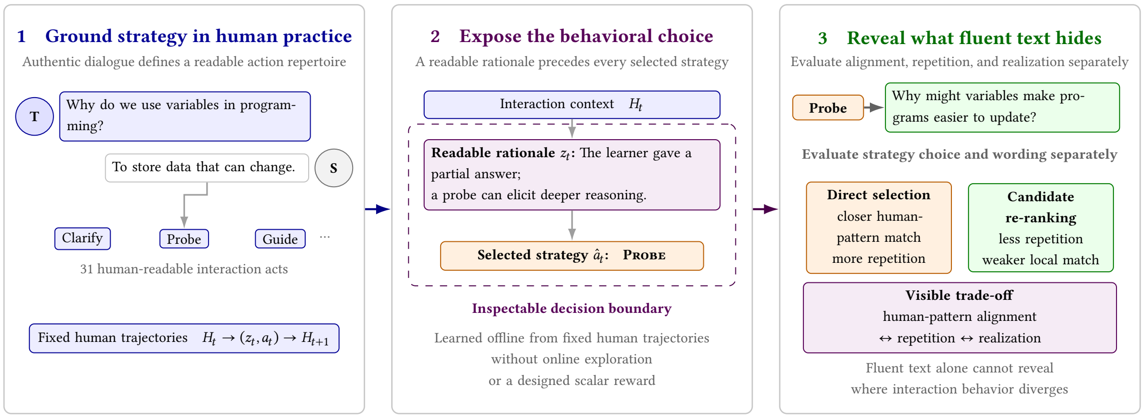

Fig. 1. **From human practice to inspectable language-mediated behavior.** Authentic dialogue grounds trajectories over readable interaction acts; LA-IQL exposes a rationale and selected strategy before language realization; and the resulting action boundary supports separate evaluation of behavioral choice and surface expression. In our classroom testbed, this boundary reveals a trade-off: direct selection more closely matches held-out human action patterns, whereas candidate re-ranking reduces repetition but moves farther from those patterns.

Human interaction unfolds through strategies such as clarifying, challenging, guiding, and building on prior contributions. Language models can generate fluent turns, yet the behavioral choices organizing their longer interactions remain implicit and need not reflect patterns found in human practice. We ask how authentic human interaction can ground strategy-rich language agents while keeping strategy selection explicit enough to evaluate. We introduce Language-Action Inverse Q-Learning (LA-IQL), an action-mediated language policy that decomposes open-ended behavior into a readable rationale, a value-based choice over human-readable interaction acts, and a separate language realization. LA-IQL learns offline from human demonstrations without online exploration or a hand-designed scalar reward. We instantiate it on cognitively annotated K–12 classroom discourse with 31 teacher–student acts and evaluate 90 formal trajectories of up to 100 turns across a matched checkpoint ablation and two inference configurations. Removing temporal-difference regularization from the evaluated Direct checkpoints increases two-step cycles from 0.1708 to 0.2607 and repeated motifs from 0.3957 to 0.5379, without a reliable change in paired bigram JSD (0.0827 vs. 0.0835). Direct action selection more closely matches held-out human action patterns than proposal-constrained candidate re-ranking (paired bigram JSD 0.0827 vs. 0.1059), but produces substantially more two-step cycles and repeated motifs. A first-order Markov control is strongest on several local alignment measures, showing that aggregate human-pattern similarity alone does not establish a context-sensitive interaction policy. In a frozen 50-state intervention, semantic context replacement changes the main policy distribution (60% action flip rate; mean JSD 0.2957), providing exploratory evidence that the policy uses dialogue text without establishing contextual appropriateness. Candidate re-ranking improves a weak automatic action–utterance consistency screen. These results show how action mediation exposes an alignment–repetition–realization trade-off that fluent utterances or aggregate similarity measures can conceal.

CCS Concepts: • **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → *Reinforcement learning*.

Additional Key Words and Phrases: human-grounded interaction, language-mediated agents, action-mediated policy, offline imitation learning, dialogue strategy

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1 Introduction

Human interaction is organized by more than locally plausible utterances. Across an extended exchange, people question, clarify, challenge, guide, take up prior contributions, and shift initiative in ways that create recognizable interaction strategies. Language increasingly mediates how interactive assistants, multi-agent systems, and embodied agents coordinate with humans and one another [Park et al. 2023; Xi et al. 2025]. For such agents, fluent wording is important, but it does not by itself establish that the interaction is behaviorally varied, responsive, or grounded in patterns of human practice.

Large language models (LLMs) commonly generate dialogues from topics, roles, or task metadata, including through role-based multi-agent simulation [Achiam et al. 2023; Kasneci et al. 2023; Park et al. 2023; Xi et al. 2025; Zhang et al. 2025]. Their behavioral choices, however, remain implicit inside open-ended text. A plausible next sentence may clarify, challenge, guide, repeat a familiar pattern, or cede initiative, and locally fluent turns can still accumulate into repetitive interaction [Li et al. 2016a; Zhao et al. 2017]. Behavioral cloning and supervised fine-tuning provide direct ways to imitate observed examples, but optimize local action or token likelihood without explicitly modeling longer interaction dynamics. Online reinforcement learning can optimize sequential behavior, but exploration with people or high-stakes simulators may be costly or unsafe.

This gap motivates our central question: *how can language agents learn human-grounded interaction strategies from authentic dialogue while keeping those strategies explicit enough to evaluate and control?* We use *human-grounded* in a deliberately bounded sense: the action repertoire, training trajectories, and comparison patterns come from observed human interaction. The term does not imply that generated dialogue is globally human-like, natural, or effective for people.

We take an action-mediated view in which a language policy first chooses a discrete, human-readable interaction act and then realizes that choice in language. This separates the source of a strategy—human demonstrations—from the wording of any individual turn. It also creates a traceable boundary at which the selected strategy and its eventual language realization can be evaluated separately. Learning from fixed trajectories avoids online exploration and a hand-designed scalar reward [Li et al. 2016b; Ouyang et al. 2022; Schulman et al. 2017].

We investigate this problem through classroom discourse, a testbed containing frequent, fine-grained interaction moves. Teacher–student exchanges are multi-role and open-ended, combining questioning, uptake, elaboration, challenge, guidance, and synthesis rather than following a fixed task script. Authentic transcripts pair this richness with an established cognitive-act annotation scheme, providing an observable operationalization of interaction strategies [Song et al. 2023]. Our aim is not to model pedagogical effectiveness or replace teachers, but to study how strategy-rich human interaction can ground a language-agent policy.

We propose **Language-Action Inverse Q-Learning (LA-IQL)**, an offline framework configured by a domain-specific action space and human demonstrations. LA-IQL learns action trajectories with rationale-conditioned offline inverse soft Q-learning over fixed transitions [Garg et al. 2021; Wulfmeier et al. 2024], selects an interaction act from

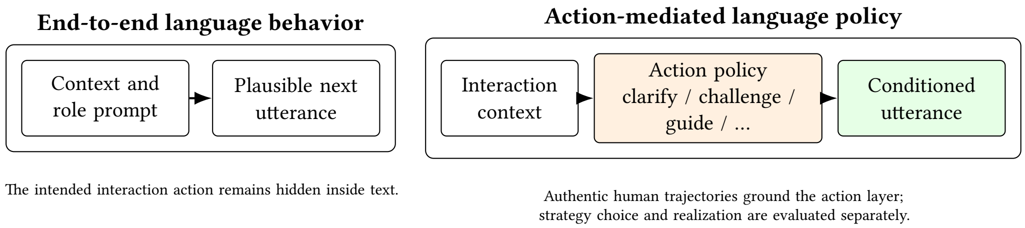

Fig. 2. Human-grounded, action-mediated interaction. LA-IQL learns a human-readable strategy layer from authentic interaction trajectories and separates strategy selection from language realization. This boundary makes behavioral patterns traceable while exposing a realization risk when the final utterance does not express the selected action.

the learned policy, and delegates its language realization to a role model. We evaluate whether generated trajectories reproduce held-out human action patterns, whether that alignment coexists with repetitive behavior, whether rationales carry action-relevant information, and whether the selected action survives language realization.

Our contributions are threefold:

- We formulate human-grounded interaction-strategy learning as an action-mediated language-policy problem, separating a human-readable behavioral choice from its surface realization. Both the action repertoire and training trajectories come from authentic human interaction, and the resulting policies are evaluated against held-out authentic human patterns.
- We instantiate this formulation in LA-IQL, a rationale-conditioned dual-head language model that combines token-level supervision, action prediction, and cross-head consistency with offline inverse Q-learning over fixed human interaction trajectories, adapting its temporal-difference regularization to a human-readable interaction-action policy.
- We show that matching local human action patterns is not sufficient evidence of context-sensitive interaction: a first-order Markov control is competitive on local alignment metrics, while the learned configurations expose an alignment–repetition–realization trade-off that aggregate similarity alone conceals.

2 Related Work

2.1 Language-Mediated Agent Interaction

End-to-end dialogue generation has progressed from sequence-to-sequence and hierarchical models to pretrained Transformers [Serban et al. 2017; Sutskever et al. 2014; Vinyals and Le 2015]. More recently, role prompting and multi-agent simulation have used LLMs to produce extended social and task interactions [Park et al. 2023; Xi et al. 2025]. Related educational systems apply supervised fine-tuning, behavioral cloning, prompting, or role simulation to teacher–student exchanges [Chu et al. 2025; Tu et al. 2023; Zhang et al. 2025]. These methods can generate fluent, role-consistent utterances, but their interaction decisions remain implicit. Consequently, surface diversity need not imply that an agent has learned the structured strategies found in human trajectories, and those strategies are difficult to control or evaluate directly [Li et al. 2016a; Zhao et al. 2017].

2.2 Proactivity, Mixed Initiative, and Agent Autonomy

HCI research treats conversational agency as more than response quality: an agent must decide when and how to contribute, how much control to assume, and how its behavior remains legible to people. Liu et al.'s Inner Thoughts framework separates covert thought formation from overt participation to support proactive conversational behavior [Liu et al. 2025]. Cheng et al. show that conversational autonomy creates non-monotonic trade-offs: full automation can improve multitasking while creating discomfort, whereas shared control can impose monitoring costs that erase performance gains [Cheng et al. 2025]. These studies motivate examining how control is allocated within a conversational system rather than assuming that either more autonomy or more user/model oversight is uniformly beneficial. Our focus is complementary: we ask how authentic human interaction can ground the action decision that precedes language realization, then compare full-action-space policy selection with a proposal-constrained division of control between the policy and role model.

2.3 Dialogue Policy Learning

Task-oriented dialogue has often been formulated as sequential decision-making, where dialogue states are mapped to abstract actions and policies are optimized with reinforcement learning [Li et al. 2016b, 2019]. Existing methods differ in state/action design, simulator construction, and online versus offline training. However, when policies operate over abstract dialogue acts, surface realization often depends on templates, retrieval, or a separate generator. This separation is problematic for classroom dialogue generation: action-level alignment may not translate into contextually appropriate utterances, especially under fine-grained interaction-act spaces. In our setting, a 31-action interaction space (two roles × 15 symbols + `END`) permits explicit action-level comparisons with expert trajectories [Amatari 2015; Song et al. 2023].

2.4 Offline Imitation for Language Models

To overcome the limitations of template-based realization and online exploration, recent work has combined language models with value-based or offline reinforcement learning [Jaques et al. 2020; Verma et al. 2022]. Model-based methods and online RL rely on accurate user simulators or human interaction, which are costly and risky in education and can induce language degradation by exploiting simulator errors [Christiano et al. 2017; Ouyang et al. 2022]. CHAI scores language-model candidates with a Q-function for task-oriented response selection [Verma et al. 2022], while IQ-Learn provides an inverse Q-learning objective for learning from expert demonstrations [Garg et al. 2021]. For language-model imitation, Wulfmeier et al. [Wulfmeier et al. 2024] reformulate inverse soft Q-learning as a temporal-difference regularized extension of MLE, making offline IRL practical for fixed supervised fine-tuning data. Our work is most closely related to these directions but differs in its target problem and modeling requirements. Rather than treating classroom response generation as the endpoint, we treat cognitively annotated classroom trajectories as a source of human-grounded interaction strategies. We use their fine-grained interaction-act space to condition policy decisions on generated rationales and to separate action-value estimation from utterance generation.

3 Method

LA-IQL treats human-grounded, language-mediated interaction as sequential decision making: at each step, the system observes dialogue history, chooses a domain-defined interaction action, and realizes that action in natural language. Following the imitation-learning perspective of offline inverse reinforcement learning, LA-IQL keeps the scalability of

Category	Symbols	Role expansion
Prior-known knowledge	Ipk, Pk	student/teacher × symbol
Personal information	Ipi, Pi	student/teacher × symbol
Analysis	Ia, An	student/teacher × symbol
Coordination	Ic, Co	student/teacher × symbol
Speculation	Is, Sp	student/teacher × symbol
Uptake	Iu, Up	student/teacher × symbol
Agreement, querying, guide	Ag, Qu, Gu	student/teacher × symbol
END	episode termination	role-independent

Table 1. Compact definition of the 31-action interaction space.

supervised fine-tuning while adding a dynamics-aware regularizer over expert trajectories. Our technical contribution is the integration of rationale-conditioned policy states, action-level TD-regularized imitation, and direct policy decoding with separate language realization rather than a new IRL objective. We additionally examine proposal-constrained candidate re-ranking as an inference-time adjustment.

3.1 Classroom Interaction as an Observable Testbed

We use a finite-context Markov decision process (MDP) approximation for offline policy learning from expert classroom demonstrations. Let H_t denote the full interaction history before step t. The policy observation $s_t = f_{\text{short}}(H_t)$ consists of the previous two dialogue turns and two action labels; it is an approximate Markov state that may omit longer-range instructional context. The action a_t is the next interaction act to be generated. It is drawn from a 31-action discrete space: {student, teacher} × 15 cognitive-act symbols (e.g., basic-knowledge questioning/responding, analysis and elaboration, summarization, transfer and innovation, response construction, agreement/challenge/guidance), plus a terminal symbol END. A selected utterance u_t and its action advance the history as $H_{t+1} = H_t \oplus (a_t, u_t)$, after which $s_{t+1} = f_{\text{short}}(H_{t+1})$; END marks episode termination. Thus the deployed transition is jointly induced by the action policy and the language realization model rather than by a_t alone. Given expert demonstrations $\mathcal{D}_E = \{(s_t, a_t, s_{t+1})\}$, we learn an action policy whose generated trajectories are evaluated for diversity and similarity to expert action patterns; language realization is evaluated separately. This action-level formulation is the first LA-IQL design choice: instead of applying inverse RL directly to open-ended tokens, we optimize over human-readable teacher–student cognitive acts and leave surface realization to a separate role model.

We evaluate interaction-pattern diversity (G1), expert-pattern alignment (G2), rationale–action consistency (G3), and automatic language-realization screening (G4).

3.2 Data Construction and Rationale Annotation

Each expert classroom trajectory is converted into fixed-window transitions. For step t, s_t contains the two most recent utterances and the two most recent action labels, a_t is the annotated next interaction act, and s_{t+1} is obtained after appending the next utterance together with a_t. We use the `deepseek-chat` API (temperature 0.7) to rewrite rationale targets for 9,168 transitions from the local state and expert next action. The artifact records the model alias and call date but not an immutable provider snapshot. The trained model does not receive the expert action as an inference-time input. It instead learns the factorization

$$p_\theta(z_t, a_t \mid s_t) = p_\theta(z_t \mid s_t)\, \pi_\theta(a_t \mid s_t, z_t). \quad (1)$$

Rationale targets that copy an action symbol or canonical label are filtered before training to reduce trivial label leakage. Automated generation checks cover length, uniqueness, and retry status; they do not constitute a semantic-quality audit. We use a fixed supervision schema:

`<Rationale>` explanation text `</Rationale>` `<Action>` role-symbol label `</Action>`.

This construction turns each expert transition into paired supervision for rationale generation and action-level policy learning without a manually specified scalar reward. Because the annotation model sees the expert action whereas the deployed rationale generator does not, training mixes teacher-forced and model-generated rationales to reduce this distribution shift.

3.3 From MLE to Expert Distribution Matching

LA-IQL augments supervised fine-tuning of a rationale-conditioned dual-head model with offline IQLearn. Whereas MLE predicts the observed action locally, the IQLearn term couples the policy to successive state values without online interaction or a learned simulator.

3.4 TD-Regularized Offline IQLearn Objective

We do not introduce a new inverse reinforcement learning objective. Instead, we adopt offline inverse soft Q-learning for language imitation [Wulfmeier et al. 2024]. Grounded in IQ-Learn [Garg et al. 2021], it recasts inverse RL as maximum-likelihood training with a temporal-difference regularizer. Our contribution is its adaptation from token-level generation to a 31-class cognitive-act policy whose Q-values are produced by a rationale-conditioned policy head. We define the augmented policy state as $x_t = (s_t, z_t)$. For teacher-forced examples, $x_t^* = (s_t, z_t^*)$ and $x_{t+1}^* = (s_{t+1}, z_{t+1}^*)$ use the corresponding annotated rationale targets; model-generated-rationale examples replace these targets with samples from $p_\theta(z \mid s)$. At inference, the language head generates z_t from s_t before action scoring. The objective contains four terms. First, $\mathcal{L}_{\text{lm}}$ is the language-modeling loss over the LLM-annotated rationale and action prefix. Second, the policy head is supervised with action-level MLE:

$$\mathcal{L}_{\text{policy}} = -\log \pi_\theta(a_t \mid x_t). \tag{2}$$

Third, the LM head induces a soft action distribution $p_{\text{LM}}(\cdot \mid x_t)$ by scoring the fixed action-label templates after the rationale. We align the policy head with this soft distribution using a KL consistency loss:

$$\mathcal{L}_{\text{consistency}} = D_{\text{KL}}(p_{\text{LM}}(\cdot \mid x_t) \parallel \pi_\theta(\cdot \mid x_t)). \tag{3}$$

Finally, following the formulation of Wulfmeier et al., the policy head produces $Q_\theta(x_t, a)$ for every $a \in \mathcal{A}$. The corresponding soft value and policy are

$$\begin{aligned} V_\theta(x_t) &= \log \sum_{a \in \mathcal{A}} \exp Q_\theta(x_t, a), \\ \log \pi_\theta(a \mid x_t) &= Q_\theta(x_t, a) - V_\theta(x_t). \end{aligned} \tag{4}$$

For an expert transition (x_t, a_t, x_{t+1}), define the action-level temporal-difference residual:

$$\delta_t := V_\theta(x_t) + \log \pi_\theta(a_t \mid x_t) - \gamma V_\theta(x_{t+1}). \tag{5}$$

We set $V_\theta(x_{t+1}) = 0$ when $a_t = \texttt{END}$. Since $V_\theta + \log \pi_\theta = Q_\theta$, the corresponding implicit reward is $\delta_t = Q_\theta(x_t, a_t) - \gamma V_\theta(x_{t+1})$. Together, $-\log \pi_\theta(a_t \mid x_t) + \lambda_4 \delta_t^2$ is the action-level counterpart of the MLE-plus-TD objective derived by

Wulfmeier et al. These terms are combined as

$$\mathcal{L} = \mathcal{L}_{\text{lm}} + \lambda_2 \mathcal{L}_{\text{policy}} + \lambda_3 \mathcal{L}_{\text{consistency}} + \lambda_4 \delta_t^2. \tag{6}$$

The coefficient λ_4 is annealed from zero after a pure-MLE warm-up, recovering action-level MLE when the TD term is disabled.

Training proceeds in two stages: we first freeze the backbone and train the policy head, pooler, and language head; we then apply LoRA fine-tuning and merge the weights. A fraction of policy-head examples uses model-generated rationales to mitigate exposure bias. Both evaluated full and no-TD checkpoints record `Qwen/Qwen3.5-0.8B` as the backbone and a 31-dimensional policy output. Their historical artifacts do not retain a complete training manifest, so optimizer settings, loss weights, training seed, and checkpoint-selection details are not reconstructed from current script defaults (Appendix B).

3.5 Rationale-Conditioned Policy Parameterization

The model uses an explicit rationale-conditioned dual-head design, which is the third LA-IQL design choice. The language head (causal LM) generates a `<Rationale>` span before the `<Action>` label; the policy head applies cross-attention pooling over the rationale span and outputs one value per action to parameterize $Q(x, a)$ and $\pi(a \mid x)$. Decisions are conditioned on the generated rationale rather than a single pooled representation.

Concretely, given an input token sequence:

- The LM head produces token-level scores for $\mathcal{L}_{\text{lm}}$ and, by scoring action-prefix templates over the 31 actions, yields a soft LM action distribution.
- The policy head pools the rationale text and outputs policy scores for $\mathcal{L}_{\text{policy}}$.
- A consistency loss $\mathcal{L}_{\text{consistency}}$ aligns the LM action-prefix distribution with the policy logits.

3.6 Direct Policy Decoding and Language Realization

At inference, LA-IQL separates short-context action selection from long-context language realization. Let $h_t = f_{\text{long}}(H_t)$ denote the role model's longer view of the same history, distinct from the policy observation s_t. The policy model first generates $z_t \sim p_\theta(z \mid s_t)$ and directly selects

$$\hat{a}_t = \arg\max_{a \in \mathcal{A}} Q_\theta(s_t, z_t, a). \tag{7}$$

The role model then produces one realization $u_t \sim q_\phi(u \mid h_t, \hat{a}_t)$. Thus the default LA-IQL inference rule exposes the full action space to the learned policy while using the longer context only to realize its selected action.

The role model is recorded as `deepseek-chat` for both teacher and student roles. It uses a long window of 30 dialogue turns and five action steps; where sampling applies, it uses a temperature of 0.8 and top-p of 1. The artifact retains the mutable API alias rather than an immutable model snapshot. In the reported evaluation, generation stops at `END` or the 100-action horizon.

We additionally explored candidate re-ranking as an inference-time adjustment. The role model samples six action–utterance candidates in a fixed schema—three for the teacher and three for the student—and thereby induces $q_\phi(\mathcal{C}_t \mid h_t)$

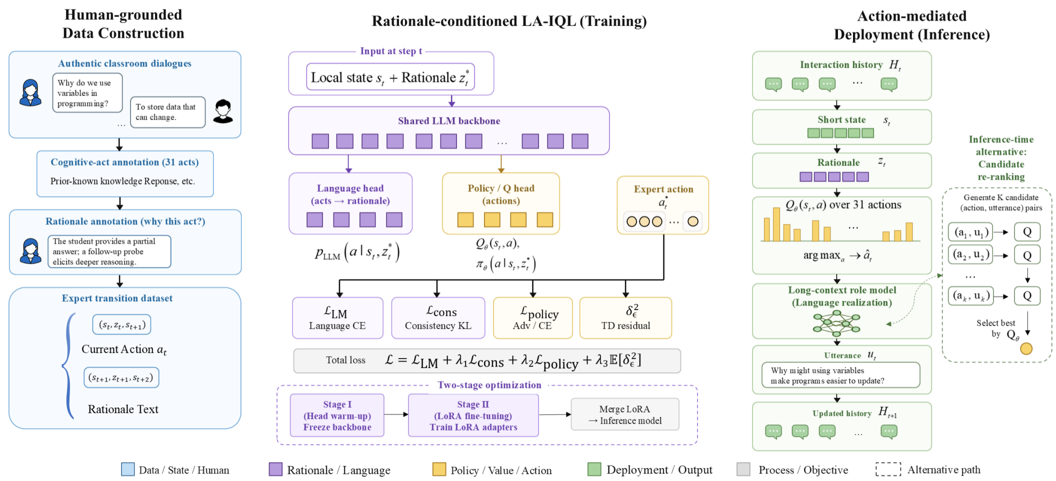

Fig. 3. **LA-IQL framework and technical flow.** Human-grounded classroom dialogue and cognitive-act annotations are converted into rationale-augmented expert transitions (left). Rationale-conditioned LA-IQL jointly trains language and policy/value components with language, consistency, policy, and temporal-difference objectives (center). At inference, the learned policy selects an interaction action and a long-context role model realizes it as an utterance; candidate re-ranking is shown as an inference-time alternative (right).

over $\mathcal{C}_t = \{(a_{t,j}, u_{t,j})\}_{j=1}^{6}$. The selected index is

$$j^* = \arg\max_{1 \le j \le 6} Q_\theta(s_t, z_t, a_{t,j}),$$
$$(a_t, u_t) = (a_{t,j^*}, u_{t,j^*}). \tag{8}$$

This adjustment scores a candidate through its associated action label and does not directly score $u_{t,j}$; candidates sharing an action are therefore indistinguishable to Q_θ, and ties retain proposal order. It is proposal-constrained because an action cannot be selected unless it appears in $\mathcal{C}_t$. Unlike default direct decoding, the role model therefore controls action support while the value model only chooses among proposed actions.

4 Experimental Design

Research questions. Our evaluation addresses four questions: **RQ1**: How do action-mediated learned policies compare with direct language generation and statistical policies in behavioral diversity, repetition, and alignment with held-out patterns from authentic human interaction? **RQ2**: How do temporal-difference regularization and the allocation of action control between direct full-action-space selection and proposal-constrained candidate re-ranking affect long-horizon behavior? **RQ3**: Do generated rationales contain action-relevant information beyond the dialogue state and explicit label leakage? **RQ4**: How reliably are selected interaction actions preserved through surface realization as classroom utterances?

Component	Recorded configuration
Policy model	`Qwen/Qwen3.5-0.8B`; 31-action rationale-conditioned dual head
Rationale annotation	`deepseek-chat`; temperature 0.7; 9,168 rewritten records
Role realization	`deepseek-chat`; temperature 0.8; top-p 1; 30-turn/5-action context
Policy observation	Two dialogue turns and two action labels; max input length 512; rationale budget 256 tokens
Formal split	35/5/10 lessons; subject–stage stratification; split seed 20260731
Rollout	10 test lessons; horizon 100; END disabled before step 50; margin 3.0
Randomness	One Full checkpoint and one no-TD checkpoint; generation seeds 42, 43, and 44

Table 2. Implementation summary. Values are transcribed from checkpoint, split, annotation, and formal-trajectory artifacts. Historical training fields not retained by those artifacts are listed as unresolved in Appendix B.

4.1 Data and Splits

We use 50 lessons from the K–12 classroom corpus described in Section 3.1. Each dialogue turn is associated with a speaker role and one cognitive-act symbol, from which we construct the 31 interaction acts and the transition tuples used by LA-IQL. We use a frozen, stratified lesson-level split of 35 training, five validation, and 10 test lessons rather than splitting by dialogue turn, so that windows from the same classroom session cannot occur in both partitions. The split is stratified by subject and school stage; lesson identifiers are disjoint across all three partitions, and two lessons used in preliminary smoke tests are excluded from the test partition. The split manifest, random seed, data hash, and split-level action frequencies are recorded before formal evaluation. The same test lessons were inspected in an earlier horizon-20 run; we disclose this prior access and do not alter test membership for the horizon-100 baseline evaluation reported below.

4.2 Baselines and Ablations

We compare LA-IQL with five completed controls: direct role-conditioned generation, a constrained teacher prompt, expert-frequency sampling, a first-order Markov policy, and uniform action sampling. The latter three isolate distributional and sequential statistics and are diagnostic controls rather than learned, state-conditioned policies. In particular, the Markov control estimates $p(a_t \mid a_{t-1})$ from the frozen training split. It is deliberately well matched to bigram- and transition-based G2 measures, but it does not observe dialogue content, the current instructional state, or a rationale, and it does not produce an utterance whose realization can be inspected. Its role is therefore to test how much local human-pattern alignment can be recovered from aggregate action transitions alone. LA-IQL uses the direct decoding rule in Equation 7. We also report *LA-IQL + candidate re-ranking*, an inference-time adjustment that substitutes the proposal-constrained rule in Equation 8 without retraining the policy. This comparison was elevated after inspection of the current held-out results and is therefore descriptive rather than a preregistered model-selection claim. Both configurations use the same checkpoint, lessons, horizon, role-model temperature, and generation seeds. However, the completed Direct trajectories use rationale temperature 0.0 whereas Candidate uses 0.6. We therefore treat them as two system configurations and do not attribute differences solely to decoding mode. To isolate the training contribution of temporal-difference regularization, we add a matched *LA-IQL w/o TD (Direct)* ablation trained without the TD term. It uses Direct decoding, rationale temperature 0.0, the same held-out lessons, role model, context windows, generation seeds, horizon, and stopping rule as the main LA-IQL system. Thus, unlike the Direct–Candidate contrast, this is a single-factor training ablation.

4.3 Evaluation Metrics

For interaction-pattern diversity (G1), the main analysis reports action Distinct-2, two-step cycle rate, empirical action entropy, and expert-pattern coverage, capturing local variety, repetitive degeneration, distributional diversity, and expert-supported breadth, respectively. For expert-pattern alignment (G2), we use Jensen–Shannon divergence between generated and expert action patterns, transition-matrix distance, and expert support for generated action n-grams. These are behavioral-distribution measures, not tests of whether an action is contextually appropriate for the dialogue state. Their overlap with the Markov control's inductive bias makes that control a demanding local-pattern reference rather than evidence of state-sensitive interaction competence. For rationale–action consistency (G3), we mask deterministic label strings and train an independent 31-way action predictor on disjoint lessons using the state, the rationale, or both. Our principal automatic measure is incremental simulatability:

$$\Delta_{\text{sim}} = F_1^{\text{macro}}(s_t, z_t \to a_t) - F_1^{\text{macro}}(s_t \to a_t), \tag{9}$$

where the target is the evaluated system's selected action. We also report masked rationale-only Macro-F1, majority, leakage, and shuffled-rationale controls. Predictability alone does not establish causal faithfulness [Hase et al. 2020; Jacovi and Goldberg 2020]; we therefore describe G3 as a simulatability diagnostic rather than an explanation-quality measure. For language realization (G4), we use a frozen automatic classifier to screen action–utterance consistency and report redundancy and rule-defined severe violations. Because this classifier has weak held-out expert performance and human judgments are incomplete, G4 does not measure contextual or classroom appropriateness; open-ended text overlap is secondary because multiple responses can be valid. Appendix D gives the complete metric and intervention protocol.

4.4 Evaluation Protocol

Action-pattern Jensen–Shannon divergence is the primary G2 automatic endpoint. Current trajectory results cover generation seeds 42–44 and 10 frozen test lessons per seed at horizon 100. These are three stochastic rollouts from each fixed checkpoint, not independently trained models. We report mean ± population standard deviation across generation-seed aggregates. For learned-system comparisons, we additionally pair the 30 matching generation-seed–lesson trajectories, cluster resampling by the 10 lessons, run 10,000 bootstrap resamples, and report percentile 95% intervals. The reported TD p-values are unadjusted within an exploratory matched-checkpoint family; the Direct–Candidate table applies Benjamini–Hochberg correction over its 11 reported comparisons. The TD ablation holds Direct decoding and rationale temperature fixed. By contrast, Direct and Candidate differ in rationale temperature and were examined after observing held-out results, so that comparison remains a descriptive system-level contrast rather than a confirmatory causal decoding effect.

Table 3 reports the core G1 metrics.

Table 4 reports G2 on the same runs. Paired JS uses the frozen 31^2 bigram vocabulary and Jeffreys smoothing ($\alpha = 0.5$); transition JS weights rows by expert frequency, and support uses training-split n-grams.

4.5 Experimental Results

Data and implementation. The corpus contains 50 lessons, 9,419 turns, 47 topics, and 9,369 transitions. We use the frozen lesson-level split described in Section 4: 35 training, five validation, and 10 test lessons. Each system has three generation runs (seeds 42–44) from the same checkpoint, with up to 100 generated actions for each test lesson. All

Method	Dist.-2 ↑	Cycle-2 ↓	H_{norm} ↑	Cov.@20 ↑
Direct Role LLM	0.4364 ± 0.0348	0.0286 ± 0.0088	0.6722 ± 0.0102	0.5500 ± 0.0408
Constrained Prompt	0.5279 ± 0.0117	0.1520 ± 0.0147	0.6818 ± 0.0015	0.5333 ± 0.0236
Uniform Random	0.9593 ± 0.0048	0.0000 ± 0.0000	0.9287 ± 0.0066	0.3500 ± 0.0000
Expert Frequency	0.6951 ± 0.0277	0.0067 ± 0.0033	0.7144 ± 0.0111	0.9500 ± 0.0000
First-order Markov	0.5144 ± 0.0072	0.0819 ± 0.0140	0.7072 ± 0.0012	1.0000 ± 0.0000
LA-IQL w/o TD (Direct)	0.4118 ± 0.0236	0.2607 ± 0.0193	0.6412 ± 0.0120	**1.0000 ± 0.0000**
LA-IQL	0.4370 ± 0.0101	0.1708 ± 0.0126	**0.6670 ± 0.0084**	**1.0000 ± 0.0000**
LA-IQL + candidate re-ranking	**0.4478 ± 0.0205**	**0.0488 ± 0.0013**	0.6664 ± 0.0131	0.7000 ± 0.0408

Table 3. Core G1 results (mean ± population SD over three generation seeds from each fixed checkpoint). Bold values mark the best learned-system result in each column; diagnostic baselines may be diverse without expert alignment.

Method	Paired bigram JS ↓	Pooled bigram JS ↓	Transition JS ↓	Bigram support ↑	Trigram support ↑
Direct Role LLM	0.1030 ± 0.0013	0.3423 ± 0.0184	0.1708 ± 0.0049	0.7532 ± 0.0167	0.4250 ± 0.0267
Constrained Prompt	0.1096 ± 0.0016	0.3795 ± 0.0072	0.1880 ± 0.0069	0.9218 ± 0.0092	0.4731 ± 0.0144
Uniform Random	0.1235 ± 0.0012	0.4661 ± 0.0042	0.2293 ± 0.0028	0.4105 ± 0.0249	0.0431 ± 0.0047
Expert Frequency	0.1036 ± 0.0006	0.2868 ± 0.0036	0.1995 ± 0.0036	0.8125 ± 0.0161	0.3316 ± 0.0032
First-order Markov	0.0791 ± 0.0009	0.0982 ± 0.0038	0.1249 ± 0.0017	0.9935 ± 0.0021	0.8557 ± 0.0169
LA-IQL w/o TD (Direct)	0.0835 ± 0.0022	**0.1530 ± 0.0177**	0.1308 ± 0.0041	0.9833 ± 0.0005	**0.8773 ± 0.0125**
LA-IQL	**0.0827 ± 0.0031**	0.1637 ± 0.0085	**0.1299 ± 0.0050**	**0.9906 ± 0.0024**	**0.8772 ± 0.0064**
LA-IQL + candidate re-ranking	0.1059 ± 0.0025	0.3116 ± 0.0132	0.1648 ± 0.0035	0.9522 ± 0.0045	0.4874 ± 0.0063

Table 4. G2 expert-pattern alignment (mean ± population SD over three generation seeds from each fixed checkpoint). Bold values mark learned-system results that are numerically best or effectively tied at the shown precision. The first-order Markov control directly models the local transitions measured by several columns but does not condition on dialogue content.

formal trajectories suppress `END` before action 50. The frequency and Markov controls use only the frozen training split. Main Direct and candidate re-ranking use the `sft_stage3/final` checkpoint; the matched training ablation uses `sft_stage3_woTD/final`. Direct action selection is the default inference rule, whereas candidate re-ranking is a proposal-constrained inference-time configuration. Main Direct and w/o-TD Direct both use rationale temperature 0.0 and otherwise identical generation settings. Candidate uses temperature 0.6, so its comparison with Direct is not a single-factor decoding ablation. Five of 30 w/o-TD trajectories terminate after 57–95 actions; all 30 main Direct trajectories reach 100 actions. This stopping behavior is an observed policy outcome under the shared rule, not a post-hoc truncation.

The evaluated TD checkpoint shows less repetitive long-horizon organization. Table 5 reports the matched Direct-decoding training ablation. Removing TD increases two-step cycles from 0.1708 to 0.2607 (TD−w/o-TD difference −0.0900, 95% CI [−0.1400, −0.0438], $p = .0002$) and motif recurrence from 0.3957 to 0.5379 (difference −0.1422, 95% CI [−0.2013, −0.0889], $p = .0002$). These correspond to 53% and 36% relative increases when TD is removed. Distinct-2 and normalized entropy are numerically higher with TD but do not differ reliably. Both systems cover all top-20 expert patterns, indicating that the reduction in recurrence is not obtained by abandoning those common patterns.

Local expert-distribution alignment changes little. Paired bigram JSD is 0.0827 with TD and 0.0835 without TD (difference −0.0007, 95% CI [−0.0063, 0.0043], $p = .8281$); transition JSD is likewise indistinguishable. Bigram support

Goal	Paired metric (TD − w/o TD)	Difference	95% CI	p
G1	Distinct-2 ↑	+0.0252	[−0.0282, 0.0688]	0.3186
G1	Cycle-2 ↓	−0.0900	[−0.1400, −0.0438]	0.0002
G1	Motif rate ↓	−0.1422	[−0.2013, −0.0889]	0.0002
G1	Normalized entropy ↑	+0.0258	[−0.0156, 0.0659]	0.2204
G2	Paired bigram JSD ↓	−0.0007	[−0.0063, 0.0043]	0.8281
G2	Transition JSD ↓	−0.0009	[−0.0091, 0.0055]	0.8581
G2	Bigram support ↑	+0.0073	[−0.00001, 0.0140]	0.0510
G4	Consistency Macro-F1 ↑	−0.0004	[−0.0025, 0.0022]	–
G4	Local trigram redundancy ↓	−0.00001	[−0.0042, 0.0041]	–

Table 5. Matched temporal-difference ablation under Direct decoding. Values compare LA-IQL with and without the TD term over the same 30 generation-seed–lesson trajectories, clustered by 10 held-out lessons, with 10,000 bootstrap resamples. Both systems use rationale temperature 0.0 and otherwise identical generation settings. Positive differences mean the TD model is numerically larger; metric arrows indicate the desirable direction.

shows a borderline descriptive advantage for TD (0.9906 vs. 0.9833, $p = .0510$). For these fixed checkpoints, the observed TD-associated difference is concentrated in recurrent sequence structure rather than local imitation fidelity. Independent training repetitions are needed to generalize this difference to the training objective.

Local transition statistics form a strong but limited reference. Tables 3 and 4 report the formal trajectory measurements. The first-order Markov control obtains the lowest paired bigram JSD (0.0791), pooled bigram JSD (0.0982), and transition-row JSD (0.1249), compared with 0.0827, 0.1637, and 0.1299 for Direct LA-IQL. It also reaches 0.9935 bigram support, while Direct LA-IQL is higher on trigram support (0.8772 vs. 0.8557).

This result follows from a substantive match between model and measurement: the control estimates $p(a_t \mid a_{t-1})$ from training trajectories, while bigram JSD and transition-row JSD measure the corresponding local action structure. High support and coverage can likewise arise from conservatively resampling training-supported transitions. These are legitimate strengths on the stated G2 endpoints, not an error to discount. However, the control never observes the dialogue text or instructional state and cannot expose whether a locally frequent transition is appropriate in context. We therefore interpret it as evidence that local human-pattern reproduction is achievable with a simple sequential statistic, and not as evidence that the Markov policy is a more capable interactive agent. This distinction motivates reporting G2 alongside state-conditioned decision traces, repetition diagnostics, and action–utterance realization checks.

A frozen intervention provides exploratory evidence of text sensitivity. We evaluated the main Full checkpoint on five pre-existing, evenly spaced states from each of the 10 held-out lessons ($N = 50$). Within each lesson, a cyclic donor replaced utterance semantics while preserving the speaker sequence, number of utterances, and action history. Relative to the original states, the probability assigned to the expert action decreased by 0.0767 (lesson-cluster bootstrap 95% CI $[0.0153, 0.1288]$), 60% of argmax actions flipped (CI $[0.48, 0.72]$), and mean policy JSD was 0.2957 (CI $[0.2476, 0.3431]$). The 31-class Macro-F1 decrease was 0.0201, with an interval crossing zero ($[-0.0185, 0.0649]$). This intervention indicates that the fixed policy distribution responds to dialogue text beyond the preserved action history. Its small sample, deterministic donor construction, and lack of human judgments mean it does not establish that the response is contextually appropriate. Appendix C records the protocol.

Within the learned architecture, alignment comes with repetitive local structure. Direct LA-IQL is closer to the held-out expert distribution than Candidate: paired bigram JSD is 0.0827 versus 0.1059, and training-expert bigram support

Goal	Paired metric (Direct − Candidate)	Difference	95% CI	BH q
G1	Distinct-2 ↑	−0.0108	[−0.0377, 0.0165]	0.5483
G1	Cycle-2 ↓	+0.1220	[0.0986, 0.1447]	< .0001
G1	Motif rate ↓	+0.2543	[0.2100, 0.2963]	< .0001
G2	Paired bigram JSD ↓	−0.0232	[−0.0305, −0.0152]	< .0001
G2	Bigram support ↑	+0.0384	[0.0283, 0.0492]	< .0001
G2	Trigram support ↑	+0.3898	[0.3551, 0.4238]	< .0001
G4	Consistency accuracy ↑	−0.0133	[−0.0233, −0.0030]	0.0140

Table 6. Paired Direct–Candidate comparison over 30 matched generation-seed–lesson units (10,000 bootstrap resamples; percentile intervals; BH correction over 11 comparisons). Positive values mean Direct is numerically larger; metric arrows indicate the desirable direction. The configurations also differ in rationale temperature, so this is a system-level comparison rather than a single-factor decoding ablation.

is 0.9906 versus 0.9522. Across 30 matched generation-seed–lesson units, the Direct–Candidate difference in bigram JSD is −0.0232, 95% CI [−0.0305, −0.0152], BH-adjusted $q < .0001$.

This alignment does not imply uniformly better interaction dynamics. Direct has substantially more two-step cycles (0.1708 vs. 0.0488; paired difference +0.1220, 95% CI [0.0986, 0.1447], $q < .0001$) and repeated motifs (0.3957 vs. 0.1413; difference +0.2543, 95% CI [0.2100, 0.2963], $q < .0001$). Distinct-2 has no reliable paired difference (−0.0108, 95% CI [−0.0377, 0.0165], $q = .5483$). Candidate therefore changes the organization of action sequences more clearly than their local combinatorial diversity.

Rationales carry action-predictive information. Under the same G3-v2 dual TF–IDF evaluator, masked rationale-only Macro-F1 is 0.3986 for Direct and 0.1013 for Candidate, compared with state-only scores of 0.0159 and 0.0123. Generation-seed-level incremental simulatability Δ_{sim} is 0.3885 for Direct and 0.0863 for Candidate. The matched w/o-TD Direct ablation obtains rationale-only Macro-F1 0.3682, combined Macro-F1 0.3629, $\Delta_{\text{sim}} = 0.3372$, and accuracy 0.6867, descriptively below the corresponding TD Direct values of 0.3986, 0.4044, 0.3885, and 0.7097. For the original Direct and Candidate configurations, the observed generation-seed-level delta exceeds all 20 within-lesson/role shuffle controls, and all 30 generation-seed–lesson deltas are positive. These results show that masked rationales contain information about the selected action. They do not show that rationales causally determine the policy decision or constitute faithful explanations; we also do not attach an inferential TD-effect claim to the descriptive G3 difference.

Language realization remains a separate risk. G4-v1 consistency accuracy is 0.0130 for Direct and 0.0263 for Candidate; the paired difference is −0.0133, 95% CI [−0.0233, −0.0030], $q = .0140$. Local trigram redundancy does not differ reliably (0.0595 vs. 0.0576; $q = .5676$). Neither configuration contains exact repetitions or hits the two rule-defined severe-violation checks. However, the frozen consistency classifier has limited held-out expert performance, and no human judgment scores are complete. We therefore interpret G4 only as evidence of a possible action–language realization gap, not as a measure of classroom quality.

The w/o-TD Direct ablation has consistency accuracy 0.0101, compared with 0.0130 for TD Direct. Their consistency Macro-F1 difference is effectively zero (TD−w/o-TD = −0.0004, 95% CI [−0.0025, 0.0022]), as is the local trigram-redundancy difference. Both have zero exact repetitions and zero rule-defined severe violations. Automatic G4 therefore provides no evidence that TD changes surface-realization quality.

Appendix A gives the full system-level comparison, G3-v2 table, and judgment status.

5 Discussion

5.1 Human-Grounded Interaction Is a Source, Not a Guarantee

Our results clarify what it means for a language policy to be human-grounded. The action vocabulary, demonstrations, and held-out comparison patterns all come from authentic classroom interaction. This grounding gives the policy a strategy repertoire that is more structured than unconstrained next-utterance generation. It does not, however, guarantee interaction that is globally human-like or contextually appropriate. The first-order Markov control's strong local alignment is the clearest counterexample: it reproduces many human-supported transitions without reading the dialogue. Human-pattern similarity is therefore evidence about a specified behavioral representation, not a complete measure of interaction quality.

This distinction matters when human traces are used to design conversational agents. Demonstrations provide both useful regularities and the limits of what the system can learn. A policy can conservatively reproduce common patterns, amplify annotation or population biases, or organize familiar actions into repetitive trajectories. Claims about human-grounded behavior should therefore identify which elements are grounded—here, the action ontology, training transitions, and evaluation reference—and which remain unverified, including experienced naturalness, pedagogical value, and responsiveness to longer-term goals.

5.2 Fixed-Checkpoint TD Differences Concentrate in Sequence Organization

The matched Direct-decoding checkpoint contrast localizes the clearest observed TD-associated difference. Removing the TD term leaves local expert-pattern alignment statistically indistinguishable, but increases two-step cycles by 53% and repeated motifs by 36%. This pattern suggests that the TD objective does not merely expand the action vocabulary or improve one-step imitation. Its observable contribution is to discourage locally plausible choices from accumulating into repetitive longer-horizon behavior. The w/o-TD policy also terminates five of 30 trajectories before the 100-action horizon, whereas the TD policy terminates none, providing further descriptive evidence of different rollout behavior. Because each condition contains one trained checkpoint, the bootstrap captures held-out lesson and generation variation but not training-seed uncertainty.

This distinction is relevant to interaction design because local appropriateness and temporal organization can fail independently. A policy may repeatedly choose actions that are individually common in human data while still producing an interaction that feels stalled. The action-mediated representation makes this failure inspectable as a sequence of recurring decisions. However, the present metrics do not establish that participants would perceive the TD trajectories as more coherent or useful; that claim requires human evaluation of complete interactions.

5.3 Action Mediation Exposes a Control Trade-off

Separating strategy selection from language realization makes visible a trade-off that end-to-end text can conceal. Direct selection gives the learned policy access to the full action space and yields stronger alignment with held-out human action patterns, but it also produces more short cycles and repeated motifs. Candidate re-ranking restricts action support to role-model proposals, reduces these repetitions, and scores higher on the automatic realization-consistency screen, but moves the resulting trajectories away from the held-out action distribution. Because the configurations also differ in rationale temperature, this is a descriptive system-level contrast rather than a causal decoding ablation. Nevertheless, it shows that increasing policy autonomy is not uniformly beneficial: control over strategy support, temporal organization, and wording must be considered together.

The explicit action boundary provides structural traceability rather than a validated user-facing explanation. It lets system designers compare the strategy selected for a state with the utterance eventually produced and localize whether a failure originates in policy choice or realization. Whether people can use this boundary effectively for oversight or intervention remains an HCI question requiring direct study.

5.4 Implications for Evaluating Strategy-Rich Language Agents

Three implications follow from the present testbed. First, action-pattern metrics should be paired with controls whose inductive biases match those metrics; otherwise, local statistical reproduction may be mistaken for context-sensitive competence. Second, evaluation should treat strategy choice and language realization as separate but coupled objects: an appropriate action label is not useful when the utterance fails to express it. Third, long-horizon repetition must be reported alongside diversity and alignment, because a policy can cover human-supported patterns while repeatedly cycling among a small subset. Together, these checks offer a more discriminating way to study language agents that aim to learn from human interaction without equating any single similarity score with human-like behavior.

6 Conclusion

We presented LA-IQL as an action-mediated language policy for learning human-grounded interaction strategies from fixed demonstrations. The approach represents strategy as a rationale-conditioned, human-readable action choice before realizing that choice with a long-context role model. A first-order Markov control matches held-out local action transitions as well as or better than the learned systems on several measures, demonstrating that aggregate human-pattern similarity alone is insufficient evidence of context-sensitive interaction. In the matched Direct-decoding ablation, temporal-difference the evaluated checkpoint trained with temporal-difference regularization shows fewer short cycles and repeated motifs without a reliable change in local expert-distribution alignment. This fixed-checkpoint result locates the main measured difference in longer-horizon behavioral organization but does not establish a training-seed-general TD effect. Within LA-IQL, direct selection is closer to the held-out human action distribution than proposal-constrained candidate re-ranking, but also repeats short cycles and motifs more often and scores lower on an automatic action–utterance consistency screen. These findings position human-grounded policy learning not as a guarantee of human-like interaction, but as a way to make strategy learning and its failures explicit. Building strategy-rich language agents requires evaluating human-pattern alignment, contextual choice, long-horizon repetition, and language realization as distinct but interdependent concerns; transfer and human-facing utility beyond this testbed remain to be established.

Limitations

LA-IQL relies on a fixed interaction-act taxonomy derived from one classroom annotation scheme. The resulting 31-action space may omit pedagogically meaningful behaviors and may not transfer unchanged across subjects, grade levels, languages, cultures, or classroom formats. Although the framework is designed around domain-specific state and action spaces and is motivated by a broader interaction-learning problem, this paper evaluates only its classroom-dialogue instantiation; transfer to conversational, multi-agent, and embodied systems remains a hypothesis to be established with domain-specific demonstrations and direct empirical evaluation. Because the method learns from offline expert demonstrations, it can reproduce annotation errors and systematic biases in the source corpus and cannot identify effective strategies that are absent from that corpus. The finite-context policy observation may miss long-range instructional goals even though the role model receives a longer dialogue history.

The completed Direct and Candidate configurations differ in rationale sampling temperature in addition to their action-selection mechanism. Their comparison therefore identifies a reproducible system-level trade-off but does not isolate the causal effect of decoding mode. A matched-temperature comparison is needed before making that attribution.

Generated rationales should not be interpreted as faithful causal explanations of the model's internal computation. They are supervised textual intermediates and can be plausible while remaining incomplete or inconsistent with the selected action. The explored proposal-constrained adjustment is limited by candidate coverage: a strong Q-function cannot select an action absent from the proposal set, and it does not distinguish utterances sharing the same action label. Direct decoding avoids this support restriction but delegates realization of its selected action to a separate role model, which may fail to express that action faithfully. Finally, automatic action-distribution metrics do not establish educational effectiveness. Claims about learning outcomes or teacher-training utility require evaluation with educators and learners in realistic settings. Moreover, the current G2 endpoints emphasize local action n-grams and transition distributions. A policy such as the first-order Markov control can score highly by conservatively reusing training-supported transitions without responding to the semantic content of the current classroom state. Our 50-state intervention provides exploratory evidence that the Full policy distribution changes with utterance semantics, but its deterministic donor and small sample do not establish contextual decision quality; broader interventions and direct state-matched judgments remain necessary.

Ethical Considerations

Classroom dialogue can contain personal information and involve minors, making privacy, consent, and data governance central to this work. Any use of classroom transcripts should follow the source dataset's consent conditions, license, and access restrictions, and should remove identifying information before model training or release. Generated dialogue must not be presented as a substitute for qualified teachers or used for high-stakes assessment without human oversight.

The system may reproduce cultural, linguistic, or pedagogical biases present in the demonstrations and in the underlying language models. In particular, a policy trained on a narrow classroom population may incorrectly treat one discourse style as universally expert. Deployment should therefore include subgroup and cross-context evaluation, educator review, clear disclosure that outputs are machine-generated, and mechanisms for users to contest or override generated recommendations. LLM-generated rationale annotations also require quality auditing because fluent explanations can conceal labeling errors or unsupported assumptions.

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Split	Lessons	Turns	Topics	Transitions
Train	35	6,352	33	6,317
Validation	5	1,039	5	1,034
Test	10	2,028	10	2,018
Total	50	9,419	47	9,369

Table 7. Corpus statistics for the frozen lesson-level split. Topic counts within rows are distinct counts and need not sum to the corpus total because one topic name occurs in more than one partition.

Variant	D2 ↑	Cycle-2 ↓	Motif ↓	JSD ↓	Cons. Acc. ↑
LA-IQL w/o TD (Direct)	0.4118	0.2607	0.5379	0.0835	0.0101
LA-IQL Direct	0.4370	0.1708	0.3957	**0.0827**	0.0130
Candidate re-ranking	0.4478	**0.0488**	**0.1413**	0.1059	**0.0263**

Table 8. Training and inference comparisons on the same held-out lessons. JSD is the G2 paired lesson-level bigram divergence; consistency accuracy is the weak G4-v1 automatic action–utterance screen.

Method	State F1	Masked R-only F1	Combined F1	Δ_{sim}	Accuracy
LA-IQL w/o TD (Direct)	**0.0257**	0.3682	0.3629	0.3372	0.6867
LA-IQL Direct	0.0159	**0.3986**	**0.4044**	**0.3885**	**0.7097**
Candidate re-ranking	0.0123	0.1013	0.0986	0.0863	0.2393

Table 9. G3-v2 rationale–action simulatability. Values are means of the three generation-seed-level scores. The target is each system's selected action; the result is an association diagnostic, not a causal-faithfulness measure.

A Additional Experimental Results

Training and inference configurations. LA-IQL directly selects the highest-valued action and then obtains one role-model realization. Candidate re-ranking instead generates six action–utterance proposals and constrains the policy to actions represented in that set. This changes the allocation of control between the policy and role model. It does not retrain the policy. The completed configurations also use different rationale temperatures (0.0 vs. 0.6), so the comparison below is a descriptive system-level contrast rather than an isolated decoding ablation. The w/o-TD Direct condition instead removes only the temporal-difference term from training and retains the main Direct generation configuration, including rationale temperature 0.0. Five of its 30 trajectories terminate after the minimum allowed horizon, compared with none for main Direct.

Figures 4 and 5 provide complementary visual summaries of these comparisons. The first emphasizes paired uncertainty, whereas the second preserves the reported metric scales and generation-seed-level variation.

G3-v2 rationale–action simulatability. The independent predictor masks action symbols, canonical names, abbreviations, and deterministic label strings. State and rationale use separate character 2–5-gram TF–IDF representations and validation-selected linear classifiers. All three systems use the same frozen evaluator protocol.

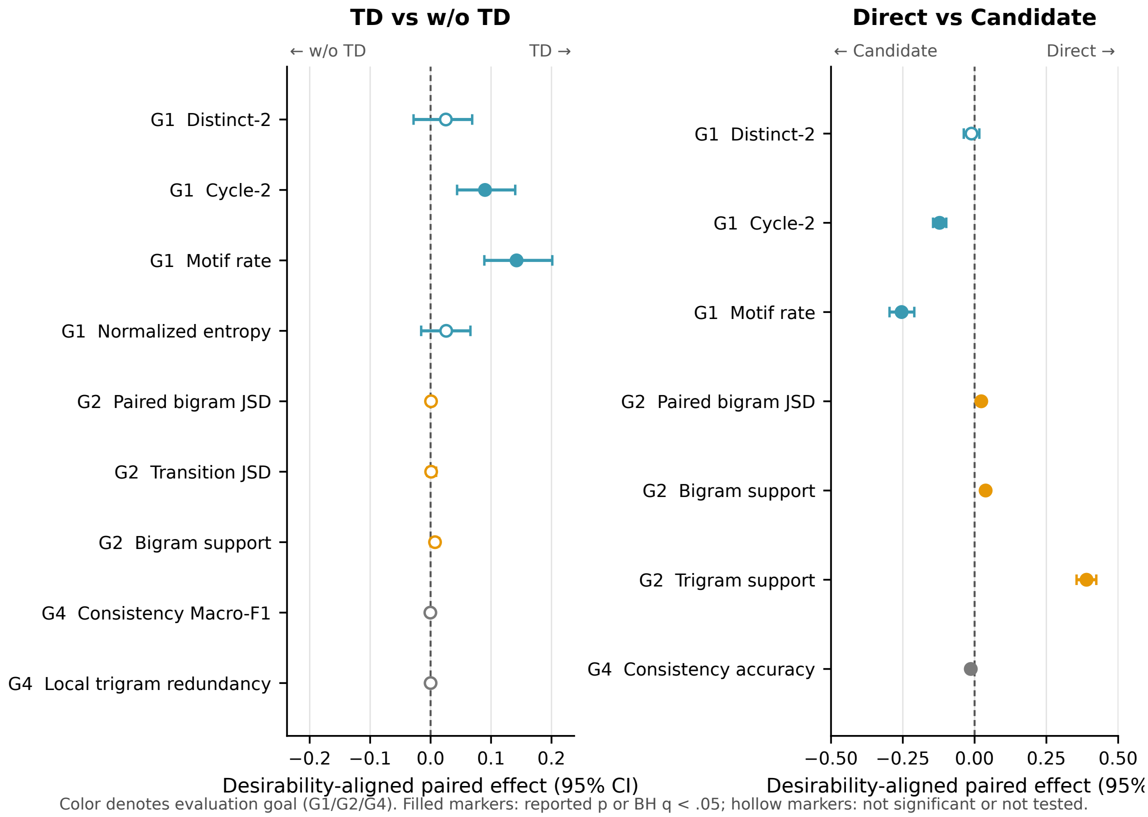

Fig. 4. Desirability-aligned paired effects for temporal-difference training and decoding configuration. For metrics marked ↓ in the result tables, the reported difference and interval are multiplied by -1 so that positive values consistently favor the first named system (TD or Direct). Horizontal bars are percentile 95% intervals from 10,000 paired bootstrap resamples clustered by the 10 held-out lessons. Filled markers denote a reported $p < .05$ or BH-adjusted $q < .05$; hollow markers denote a non-significant or untested comparison. The Direct–Candidate panel is a descriptive system-level comparison because rationale temperature also differs between configurations.

Judgment status. Blinded items, randomized manifests, and the rating rubric have been generated, but no completed human judgment scores are present. We therefore report only the automatic G4-v1 screen and make no claim about factuality, safety, pedagogical appropriateness, experienced naturalness, or learning outcomes from the zero severe-rule violation rate.

B Implementation and Reproducibility

Evidence hierarchy and unresolved training fields. We transcribed inference settings from the terminal `config` object in the formal trajectory JSON files, data provenance from the frozen split manifest, and model identity from checkpoint metadata. These artifacts are authoritative for the reported runs. The current training scripts have changed since the main Full and no-TD checkpoints were produced, while their checkpoint metadata retains only the backbone and policy dimension. Consequently, the historical optimizer, learning rate, batch size, epoch count, loss weights, teacher/self-generated rationale ratio, training seed, initialization, code revision, and checkpoint-selection rule are not recoverable

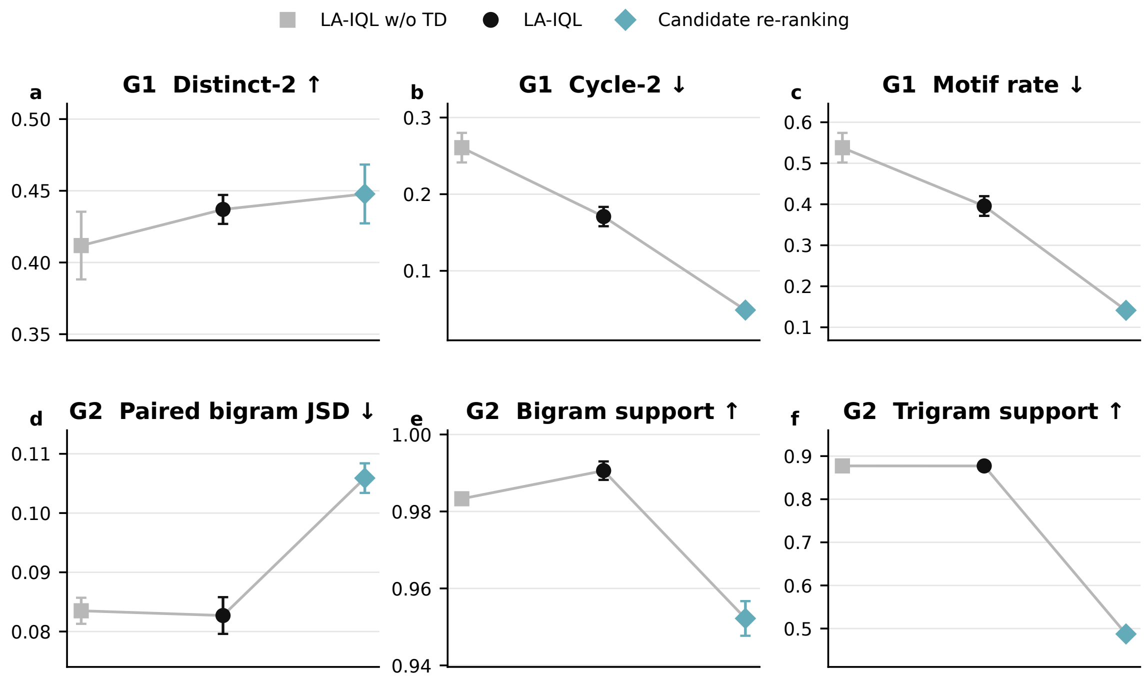

Fig. 5. Core G1 and G2 results for the three learned-system configurations. Points show means and error bars show population standard deviations across generation seeds 42–44 from each fixed checkpoint, with 10 held-out lessons per seed. Connecting lines are visual guides rather than temporal trajectories; arrows indicate the desirable metric direction.

System	Checkpoint / objective	TD	Selection	Rat. temp.	Role proposals	Randomness
Full Direct	`sft_stage3/final`; four-term objective	on	full-space argmax	0.0	one realization	gen. 42–44
Full Candidate	same Full checkpoint	on	six-proposal re-rank	0.6	3/role	gen. 42–44
No-TD Direct	`sft_stage3_woTD/final`; TD removed	off	full-space argmax	0.0	one realization	gen. 42–44
Frozen BC	`sft_bc_seed*`; next-action CE	off	full-space argmax	n/a	n/a	train 42–44
LoRA BC	`lora_policy_seed*`; next-action CE	off	full-space argmax	n/a	n/a	train 42–44

Table 10. System provenance matrix. “gen.” denotes generation seeds applied to a fixed checkpoint; “train” denotes independently trained BC checkpoints. The BC models are completed diagnostics but are not included in the formal horizon-100 trajectory tables.

from the active artifacts. We neither substitute current defaults nor describe the no-TD result as replicated across training seeds.

Completed BC training metadata. The frozen-backbone BC checkpoints use `Qwen/Qwen3.5-0.8B`, 6,181 training samples, next-action cross-entropy, three epochs, batch size 4, learning rate 2×10^{-5}, and maximum length 512. The LoRA BC checkpoints use the same backbone, objective, and training set plus 999 validation samples; they use three epochs, batch size 32, learning rate 10^{-4}, maximum length 256, rank 8, bfloat16, and one gradient-accumulation step. Each family retains independent training seeds 42–44 and per-epoch metadata.

Artifact	Recorded provenance
Split	`g1-formal-v1-deepseek-rationale-rewrite-v1`; seed 20260731; lesson unit; subject–stage strata
Source data	source SHA-256 prefix `5c76ac1d4c2f`; original `e92eef3da25a`; rewritten `b18904625ef0`; expert `7bda9edd0e07`
Rewrite	`rationale-rewrite-v1`; run prefix `bd7188c66f9a`; 9,168 records in 50 files; 8,786/380/2 records accepted on attempts 1/2/3
Full checkpoint	`Qwen/Qwen3.5-0.8B`; 31 outputs; checkpoint SHA-256 prefix `a68c26bddc63`
No-TD checkpoint	same backbone/output size; checkpoint SHA-256 prefix `ed0b35180b48`
External models	rationale rewrite: `deepseek-chat`, 0.7, 2026-08-09; role realization: `deepseek-chat`, 0.8

Table 11. Data, annotation, and checkpoint provenance. Hash prefixes are shown for readability; the machine-readable manifests retain the full SHA-256 values. The external API records contain aliases, not immutable provider revisions.

Endpoint	Original	Counterfactual	Difference / shift
Accuracy	0.4000	0.3000	−0.1000
Macro-F1 (31)	0.1486	0.1285	−0.0201
Expert-action probability	0.2732	0.1965	−0.0767
Prediction flip rate	—	—	0.6000
Policy JSD	—	—	0.2957

Table 12. Exploratory context intervention for the main Full checkpoint ($N = 50$ states, 10 lessons). Lesson-cluster intervals appear in the main text. The intervention tests distributional sensitivity, not contextual correctness.

Inference configuration and quality gates. All reported Full, Candidate, and no-TD formal trajectories use a two-turn, two-action policy window; a 30-turn, five-action role window; policy maximum length 512; rationale budget 256 new tokens; horizon 100; role temperature 0.8; three candidates per role where candidate re-ranking applies; and `END` suppression through step 49 with logit margin 3.0. The Full and no-TD generation gates each contain 50 samples and report 100% structural validity, closed rationale/action tags, no degeneration, and no length-limit hits. These checks establish schema validity, not policy or rationale quality.

Software environment. The locked environment requires Python 3.10 or newer and resolves PyTorch 2.11.0+cu128, Transformers 5.14.1, PEFT 0.20.0, scikit-learn 1.7.2, SciPy 1.15.3, OpenAI 2.52.0, and Google GenAI 2.16.0. The dependency lockfile and CUDA 12.8 wheel index are part of the code artifact.

C Context-Sensitivity Intervention

The frozen `context-sensitivity-v1` protocol selects five pre-existing, evenly spaced held-out states per lesson from the frozen G4 candidate pools. For each state, it constructs a within-lesson cyclic semantic donor while preserving the number and sequence of speakers and the complete action history. Rationales are greedily decoded with temperature 0 and a 160-token budget; `END` suppression is not applied because this is a state-level test. Intervals use 10,000 percentile bootstrap resamples clustered by lesson with seed 20260822.

Method	Runs	Dist.-1	Dist.-2	Dist.-3	Motif ↓	Cycle-2 ↓	Cov.@20	H_{norm}
Direct Role LLM	10×3	0.2692 ± 0.0050	0.4364 ± 0.0348	0.5913 ± 0.0427	0.4486 ± 0.0287	0.0286 ± 0.0088	0.5500 ± 0.0408	0.6722 ± 0.0102
Constrained Prompt	10×3	0.2866 ± 0.0029	0.5279 ± 0.0117	0.7264 ± 0.0144	0.3403 ± 0.0204	0.1520 ± 0.0147	0.5333 ± 0.0236	0.6818 ± 0.0015
Uniform Random	10×3	0.4149 ± 0.0198	0.9593 ± 0.0048	0.9986 ± 0.0013	0.0048 ± 0.0012	0.0000 ± 0.0000	0.3500 ± 0.0000	0.9287 ± 0.0066
Expert Frequency	10×3	0.2154 ± 0.0145	0.6951 ± 0.0277	0.9449 ± 0.0069	0.0795 ± 0.0098	0.0067 ± 0.0033	0.9500 ± 0.0000	0.7144 ± 0.0111
First-order Markov	10×3	0.2004 ± 0.0047	0.5144 ± 0.0072	0.7815 ± 0.0026	0.2136 ± 0.0417	0.0819 ± 0.0140	1.0000 ± 0.0000	0.7072 ± 0.0012
LA-IQL w/o TD (Direct)	10×3	0.1606 ± 0.0088	0.4118 ± 0.0236	0.6352 ± 0.0219	0.5379 ± 0.0361	0.2607 ± 0.0193	1.0000 ± 0.0000	0.6412 ± 0.0120
LA-IQL	10×3	0.1663 ± 0.0034	0.4370 ± 0.0101	0.6983 ± 0.0050	0.3957 ± 0.0240	0.1708 ± 0.0126	1.0000 ± 0.0000	0.6670 ± 0.0084
LA-IQL + candidate re-ranking	10×3	0.1493 ± 0.0070	0.4478 ± 0.0205	0.7221 ± 0.0238	0.1413 ± 0.0017	0.0488 ± 0.0013	0.7000 ± 0.0408	0.6664 ± 0.0131

Table 13. Full G1 statistics for all completed systems. Values are mean ± population standard deviation across generation seeds 42, 43, and 44; each seed contains 10 held-out lessons. Learned systems reuse one checkpoint per condition across those seeds.

D Evaluation Details

The complete G1 suite additionally includes Distinct-1/3, repeated local motifs, and cross-trajectory variance. For G3, masking removes action symbols, canonical names, abbreviations, and deterministic label strings before prediction. We report leakage rates, per-class scores, confusion matrices, and majority and shuffled-rationale controls.

The current G3 evaluation establishes association through independent prediction and within-lesson/role shuffle controls. Remove, paraphrase, and role-compatible counterfactual interventions [DeYoung et al. 2020] remain future tests of causal sensitivity. Blinded action-support and classroom- appropriateness materials have been prepared, but no completed human judgment scores are reported. We therefore avoid claims about explanation faithfulness, educational effectiveness, or experienced naturalness.

E Full G1 Statistics